Nagizaaz Shaik

LLM & GENERATIVE AI ENGINEER · NLP · MODEL EVALUATION · AGENTIC AI

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LLM/NLP engineer who improves models by experiment: five years in data and ML engineering, two of them shipping production LLM systems in regulated finance. I test models on held-out data, fine-tune classifiers and ship what the data supports.

SKILLS

LLMs & NLP: Claude, OpenAI, Qwen2.5, Hugging Face, BERT, LoRA / PEFT, text classification, distillation, vLLM, Ollama
Experimentation & Evaluation: held-out tests, paired bootstrap CIs, ablations, A/B tests, calibration, RAGAS, LLM-as-judge
Agentic AI & RAG: LangGraph, CrewAI, MCP, tool calling, agent memory, LangChain, Pinecone, FAISS, hybrid search, reranking
Stack: Python, SQL, Go · PyTorch, scikit-learn, XGBoost, MLflow, SageMaker, FastAPI, Docker, Kubernetes, Airflow

EXPERIENCE

Doctoral Researcher, Applied AI | Belhaven University, Jackson, MS — Aug 2026 – Present

- To measure how far a known AI-agent sandbox bypass had spread, traced the copied code into 25 open-source projects; 23 were still unpatched, and the report earned reporter credit on GHSA-m5v3-6ccr-cfgx (working paper *When Vendoring Defeats the Patch*).
- To test whether small financial text classifiers can be manipulated, ran planted-label experiments in five markets with controls and three seeds; a planted label pushed a LoRA model to the attacker's label on up to 92% of items (*Fragile in Different Ways*).

ML Platform Engineer · Contract | Wright-Patt Credit Union ($9B assets), Fairborn, OH — Jul 2025 – Jun 2026

- To give teams a governed way to use LLMs, built an internal gateway over Claude, OpenAI, and Ollama with evaluation guardrails, cost controls, and RAGAS grounding checks: one approved GenAI path for the 12–15-person IT team.
- As the founding ML platform engineer, built the first AWS ML platform (SageMaker endpoints, MLflow registry with gated promotion and automated rollback, VPC isolation), cutting model deployment failures 85% in 60 days.
- Designed security for all ML workloads (AES-256, least-privilege IAM, audit logging); the platform passed two consecutive NCUA audits with zero findings. Mentored two engineers into ML and platform ownership.

Machine Learning Engineer · Contract | Day Air Credit Union ($830M assets), Kettering, OH — Jul 2024 – Jun 2025

- Compliance staff spent about two hours per regulatory question; built RAG over 500+ policy and NCUA documents (LangChain, vector search, Claude), cutting research to under 20 minutes per query.
- To fix ticket misrouting, fine-tuned a 14-class BERT intent classifier (FastAPI, sub-200 ms p99) routing 10,000+ tickets a week.
- To automate first-pass underwriting, shipped an XGBoost credit-risk model scoring 500+ loan applications a day, with weekly retraining gated on AUC, precision, and calibration; drift monitoring caught two drift events before they reached credit decisions.

Senior Data Engineer | Cognizant (Financial Services client), Hyderabad, India — Jul 2020 – Jun 2023

- Analytical queries on a legacy Oracle warehouse were slow for five business units; led the migration to Redshift, rebuilding 20+ stored procedures as Airflow DAGs and cutting query runtime 60%.
- Built 15 production ETL/ELT pipelines with multi-stage validation that surfaced a silent upstream corruption affecting three months of financial reports; promoted from Associate to Senior in two years.

Python Intern | Uber, Hyderabad, India — Jan 2019 – Oct 2019

PROJECTS

Financial Text Classification with LoRA | PyTorch, PEFT, Qwen2.5 — github.com/shaikn6/fin-lora

A LoRA-tuned Qwen2.5-0.5B trained on tweets scored only 59.0% on news; retrained on news, tweets and headlines, it reached 84.7% vs 71.7% for TF-IDF and 73.6% for FinBERT. A confidence cascade then cut average latency 63% for 0.4 points of accuracy.

lethe: Agent Memory Server | Go, Python, LangChain, MCP — github.com/shaikn6/lethe

Memory server for AI agents whose ranking is learned from feedback. The first learner lost to the hand-set rule (0.611 vs 0.641 evidence in the top 10); adding forced exploration fixed it: 0.707 on held-out LoCoMo conversations.

Credit-Risk Model Benchmark | Python, scikit-learn — github.com/shaikn6/credit-arena

Compared six model families on 30,000 accounts with paired bootstrap tests: gradient boosting beat logistic regression by +0.0275 AUC (95% CI 0.019–0.037) and cut expected cost about 8%; nothing beat it significantly.

Open Source | 15 merged PRs: crewAI (60K stars, incl. a fix for a bug I reported), Ray (44K), MLflow (28K) — github.com/shaikn6

EDUCATION & CERTIFICATIONS

Belhaven University — Doctor of Business Administration (DBA), Applied AI — Aug 2026 – Present

University of Dayton — M.S. Computer Science · GPA 3.93 / 4.0 — Aug 2023 – May 2025

CMR Engineering College — B.Tech. Computer Science and Engineering · GPA 8.9 / 10 — Aug 2016 – May 2020

AWS Certified Machine Learning – Specialty • AWS Certified Data Engineer – Associate • GCP Professional Data Engineer